

Adjunct Island Effect on Wh-Scrambling in Korean

Jiayi Lu
Northwestern University
jiayi.lu@northwestern.edu
ORCID ID: 0000-0002-3314-5666

Nayoun Kim
Sungkyunkwan University
nayoun@skku.edu

Abstract In this squib, we present experimental evidence that Korean *wh*-scrambling, unlike non-*wh*-scrambling or *wh*-in-situ, is constrained by the *-ttay* temporal adjunct island effect. We further show that this adjunct island effect cannot be reduced to a discourse-level constraint on question formation that bans new-information-seeking *wh*-questions, formed with or without movement, targeting discourse backgrounded information: *-ttay* adjunct clauses do not constrain in-situ *wh*-DPs even though they are interpreted as discourse backgrounded. We conclude that the island effect observed on *wh*-scrambling instead reflects a constraint on genuine syntactic movement. Our finding supports a movement-based analysis of long-distance *wh*-scrambling in Korean, and points to the possibility that the scrambling of *wh*-elements and non-*wh*-elements may be derived via different mechanisms.

Keywords: island effect, *wh*-scrambling, *wh*-in-situ, experimental syntax

1 Introduction

Scrambling refers to the phenomenon whereby, in some languages such as Japanese and Korean, constituents in sentences can occur in various different orders that deviate from the canonical one (Ross 1967). For example, in the Korean sentence (1a), the object DP precedes the subject, deviating from the canonical SOV word order. Scrambling may also be cross-clausal in Korean (i.e., *long-distance scrambling*): in (1b), the embedded object precedes the matrix subject.

- (1) a. Eunji-lul Hyelini-ka mannass-ta.
Eunji-ACC Hyelin-NOM meet.PST-DECL
'Hyelin met Eunji.'
- b. i chaek-lul Cheolsu-nun Eunji-ka Hyelini-eykey cwuess-ta-ko
This book-ACC Cheolsu-TOP Eunji-NOM Hyelin-DAT give.PST-decl-COMP
malhayss-ta.
say.PST-DECL
'Cheolsu said that Eunji gave this book to Hyelin.'

Long-distance scrambling also applies to *wh*-elements in Korean. Since Korean is a *wh*-in-situ language, *wh*-elements do not need to move in the formation of *wh*-questions, as shown in example (2a). Nevertheless, as shown in (2b), *wh*-elements can also be fronted through long-distance scrambling, resulting in a surface form similar to *wh*-questions in *wh*-movement languages. We shall call this operation *wh*-scrambling.

- (2) a. Cheolsu-nun Eunji-ka mwues-ul ilkess-ta-ko malhayss-ni?
Cheolsu.TOP Eunji.NOM what.ACC read.PST-decl-COMP say.PST-Q
'What did Cheolsu say that Eunji read?'

- b. mwues-ul Cheolsu-nun Eunji-ka ilkess-ta-ko malhayss-ni?
 what.ACC Chelswu.TOP Eunji.NOM read.PST-decl-COMP say.PST-Q
 ‘What did Cheolsu say that Eunji read?’

Long-distance scrambling in Korean has received two main classes of analyses: base-generation analyses, and movement analyses. Under the base-generation analysis, the scrambled elements are merged directly at their surface positions (Kornfilt & Bayer 1994; Neeleman 1994; Bošković & Takahashi 1998; Fanselow 2001: *inter alia*). For example, under Bošković & Takahashi’s (1998) proposal, the scrambled element (e.g. ‘this book’ in (1b)) is base-generated in its surface TP-adjoined position (and stays there for PF purposes). Since the scrambled element cannot be θ -marked in its surface position, as a last resort, it undergoes an LF lowering operation to receive its θ -role from the embedded V.

Under the movement-based analyses, scrambling involves movement of the scrambled element from its theta-position to its surface position (Ross 1967; Chomsky & Lasnik 1977; Saito 1982; 1985; Hoji 1985; Kuroda 1988; Fukui 1993; Lee 1993; Miyagawa 1997; Bailyn 2001; Ko 2007; 2014: *inter alia*).

The base-generation accounts, unlike the movement-based accounts, makes room for the possibility that scrambling may not be island sensitive: it is possible that the operations to get a base-generated scrambled DP θ -marked (e.g., LF lowering) may not respect the same locality constraints that restrict overt A’-movements. In fact, in Korean, it has been repeatedly shown that long-distance scrambling can escape various island domains. Ko et al. (2019) tested both leftward and rightward scrambling in Korean experimentally, and showed that both are insensitive to the adjunct and *wh*-islands; Kim & Ji (2023) showed that Korean scrambling is insensitive to the Complex-NP island; Kim (2026) showed that Korean scrambling shows no adjunct island effect, but a statistically significant yet non-robust relative clause island effect. These findings largely contrast with the island sensitive nature of other A’-movements in Korean such as relativization and topicalization (Han 1992; Na & Huck 1993; Han & Kim 2004; Han 2013; Kim 2019; cf. Kim 2013), and some may consider these findings evidence against analyzing scrambling in Korean as a movement operation.

In this paper, we further examine the nature of Korean long-distance scrambling. Specifically, we present a piece of unexpected finding that goes against the previous generalization that Korean scrambling is island insensitive: we find that the scrambling of *wh*-DPs in Korean is restricted by temporal clausal adjuncts headed by *-ttay* (when). This contrasts with previous experimental work by Ko et al. (2019) and Kim (2026) that adopt similar factorial designs to probe for island effects, both of which reported no island effect for the scrambling of non-*wh* DP out of clausal adjuncts in Korean. Our finding supports the analysis that the long-distance scrambling of *wh*-elements involves regular upward syntactic movement in Korean (contra Bošković & Takahashi’s 1998 analysis for Japanese); it also points to the possibility that *wh*-scrambling and the scrambling of non-*wh* DP may be derived via different mechanisms, given the different findings in the current study and Ko et al. (2019) and Kim (2026): non-*wh*-scrambling may be derived via base-generation of the seemingly displaced elements in the left periphery, whereas this base-generation option is not available for *wh*-elements.

In this paper, we present results from three experiments. Experiment 1 presents evidence for the adjunct island effect on *wh*-scrambling in Korean. Experiment 2 shows that the clausal adjuncts tested in Experiment 1 are interpreted as discourse backgrounded. Experiment 3 shows that adjunct islands do not restrict in-situ *wh*-DPs, in line with previous generalizations that in-situ *wh*-DPs are not restricted by island effects in Korean (Kim & Goodall 2016; Kim et al. 2023, cf. Cho 2018) and crosslinguistically (Huang 1982;

Tsai 1994;  Jin & Yan 2025; Tian et al. 2022; Sprouse et al. 2011; Çakır 2016; 2017; Tsai 2009, cf. Lu et al. 2020; Tanaka 2025; Kim et al. 2023; Yue & Xiang 2025; Chen 2024; Chieu 2022). Taken together, Experiments 2 and 3 confirm that the results of Experiment 1 reflect a genuine syntactic island effect on movement, as opposed to a discourse-level ban on new-information-seeking questions targeting discourse-backgrounded information (Abeillé et al. 2020; Goldberg 2013; Namboodiripad et al. 2022; Gunco & Goldberg 2022; 2023; Lu et al. 2025; Ambridge & Goldberg 2008).

2 Experiment 1

2.1 Methods, materials, and participants

Previous experimental studies have shown that the scrambling of non-*wh* DPs in Korean is not sensitive to the adjunct island effect (Ko et al. 2019; Kim 2026). In this experiment, we test whether *wh*-scrambling is restricted by the adjunct island effect. Specifically, we examine whether *wh*-scrambling is restricted by temporal adjunct clauses.¹

The experiment uses a formal acceptability judgment paradigm (Sprouse et al. 2012; 2016; Kush et al. 2019). Example stimuli are shown in (3). All stimuli are presented as question-answer pairs. The question sentence comes in four different conditions yielded by a 2-by-2 factorial design manipulating two factors: STRUCTURE of the embedded clause (island vs. non-island) and LENGTH of the dependency between the *wh*-element and its θ -position (short vs. long). The structure of the embedded clause was manipulated to be either an temporal adjunct clause (island) or a complement clause (non-island), and the fronted *wh*-element was manipulated such that its θ -position is either in the matrix clause (short dependency) or in the embedded clause (long dependency). All four conditions were lexically matched except for the matrix verb, which necessarily differed between complement-clause and adjunct-clause conditions for selectional reasons (e.g., ‘hear’ in complement-clause conditions and ‘laugh’ in adjunct-clause conditions).

Each question sentence is presented along with an answer in the form of a proper name. This makes sure that the participants would access the intended reading of the question sentence as a matrix *wh*-question, and the *wh*-element would not be interpreted as an indefinite yielding an unintended polar question interpretation. For example, without the answer sentence, some speakers may interpret the question sentence in (3a) as a polar question meaning “*Did Minho laugh when Jisoo met someone?*”. This question-answer pair stimuli format follows the design of previous studies that examined island-sensitivity of *wh*-questions in Korean (Kim & Goodall 2016). Participants were instructed to read the entire dialogue and then rate the acceptability of the question sentence given the dialogue between the question and the answer using a 1-7 Likert scale. A total of 60 self-reported native speakers of Korean were recruited for the experiment and were compensated approximately \$3 for their participation. One participant was excluded from analysis because the 95% confidence intervals of their responses to the grammatical and ungrammatical fillers overlapped. Each participant was presented 16 critical items in a Latin Square fashion, along with 20 filler items.

¹ The choice of testing temporal adjunct clauses over other types of adjunct clauses (e.g., causal and conditional adjunct clauses) in Experiment 1 provides maximal comparability with previous experimental work on Korean that also used temporal adjunct clauses to test for the adjunct island effect (e.g., Kim & Goodall 2016; Ko et al. 2019). Nevertheless, we should note that different types of adjunct clauses may have different islandhood status (Bondevik et al. 2021; Bondevik & Lohndal 2023; Truswell 2007; *inter alia*). The claim made in the current study is specific to the *-ttay* temporal adjunct island effect on *wh*-scrambling.

Following previous work (Sprouse et al. 2012; 2016; Kush et al. 2019), we probe for the temporal adjunct island effect as the interaction of STRUCTURE and LENGTH: if *wh*-scrambling is sensitive to the temporal adjunct island, the length penalty (i.e., the contrast between short and long *wh*-dependencies) should be greater when it spans an adjunct clause compared to when it spans a complement clause.

(3) Experiment 1 sample stimuli sentences

a. *Island, Long*

cilmwun: nwukwu-lul Minhoo-nun Jisoo-ka mannassul-ttay wus-ess-ni?
 Question: Who-ACC Minhoo-TOP Jisoo-NOM meet.PST.ADN-when laugh-PST-Q?
 tap: Hyunwoo
 Answer: Hyunwoo
 ‘Q: Who did Minhoo laugh when Jisoo met? A: Hyunwoo’

b. *Island, Short*

cilmwun: nwu-ka Jisoo-ka Hyunwoo-lul mannassul-ttay wus-ess-ni?
 Question: Who-NOM Jisoo-NOM Hyunwoo-ACC meet.PST.ADN-when laugh-PST-Q?
 tap: Minhoo
 Answer: Minhoo
 ‘Q: Who laughed when Jisoo met Hyunwoo? A: Minhoo’

c. *Non-island, Long*

cilmwun: nwukwu-lul Minhoo-nun Jisoo-ka mannass-ta-ko tul-ess-ni?
 Question: Who-ACC Minhoo-TOP Jisoo-NOM meet.PST-DECL-COMP hear-PST-Q?
 tap: Hyunwoo
 Answer: Hyunwoo
 ‘Q: Who did Minhoo hear that Jisoo met? A: Hyunwoo’

d. *Non-island, Short*

cilmwun: nwu-ka Jisoo-ka Hyunwoo-lul mannass-ta-ko tul-ess-ni?
 Question: Who.NOM Jisoo-NOM Hyunwoo-ACC meet.PST-DECL-COMP hear-PST-Q?
 tap: Minhoo
 Answer: Minhoo
 ‘Q: Who heard that Jisoo met Hyunwoo? A: Minhoo’

2.2 Results and discussion

The mean acceptability rating for each condition is shown in Figure 1. We fit an ordinal regression model predicting the acceptability rating with the sum-coded fixed effects of STRUCTURE, LENGTH, and their interaction. Also included were the by-item random intercept and slopes for all the fixed effects, the maximal random-effect structure allowing model convergence. There is a significant main effect of STRUCTURE ($\beta = 0.16$, $SE = 0.079$, $z = 2.04$, $p < 0.05$) where the island conditions are rated lower than the non-island conditions, and a significant main effect of LENGTH ($\beta = 2.01$, $SE = 0.020$, $z = 10.03$, $p < 0.001$) where the short conditions are rated higher than the long conditions. There is also a significant STRUCTURE $\times$ LENGTH interaction ($\beta = -0.30$, $SE = 0.069$, $z = -4.29$, $p < 0.001$), where the length effect is greater for the island conditions than for the non-island conditions. The significant interaction term suggests that there is a temporal adjunct island effect.

This result crucially contrasts with the findings of Ko et al. (2019) and Kim (2026), where no adjunct island effect is found for the long-distance scrambling of non-*wh* DPs. One would be tempted to conclude that the findings of this experiment favors a movement-

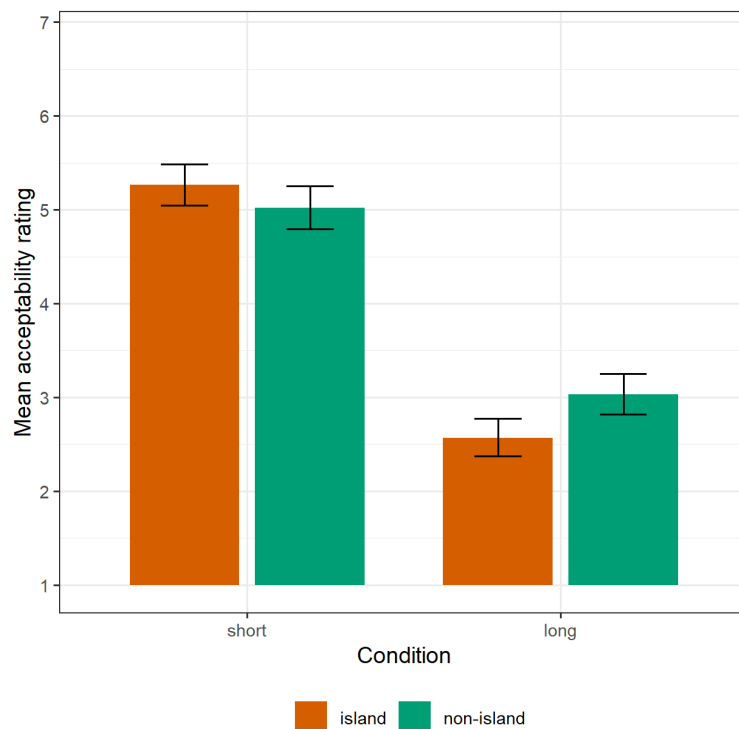


Figure 1: Acceptability ratings for Experiment 1

based analysis of *wh*-scrambling over a base-generation analysis.² But there is a potential confound: one possible explanation for the difference is that the effect found in the current experiment is in fact not a genuine syntactic island effect on movements, but rather a discourse-level penalty unique to *wh*-questions. Some suggested that forming *wh*-questions targeting discourse backgrounded information results in degradedness due to the clash between the fact that *wh*-questions should be new-information-seeking, and the discourse-old status of backgrounded and presupposed information (Erteschik-Shir 1973; Ambridge & Goldberg 2008; Goldberg 2013; Namboodiripad et al. 2022; Cuneo & Goldberg 2022; 2023; Lu et al. 2025). In the context of the temporal adjunct island effect, a discourse-based explanation goes as follows. Let's assume, for the moment, that temporal adjunct clauses are interpreted as discourse backgrounded in Korean (we will explicitly verify this assumption in Experiment 2). By virtue of using a temporal adjunct clause in a sentence, the speaker is signaling that the content embedded inside the adjunct clause is discourse backgrounded. At the same time, by virtue of uttering a new-information-seeking *wh*-question, the speaker signals that the questioned content is foregrounded and at-issue. Under this analysis, *wh*-questions, regardless of whether there is genuine syntactic movement of the *wh*-elements, cannot target elements contained inside clausal adjuncts: doing so would give rise to a clash in the discourse-backgrounding function of the adjunct clause and the discourse-foregrounding function of new-information-seeking questions. By contrast, forming questions targeting non-backgrounded constituents (e.g., complement clauses) would not give rise to this clash in discourse function.

To test whether the temporal adjunct island effect we observed reflects a movement constraint or a discourse-level constraint, we further conducted Experiments 2 and 3 to test for the temporal adjunct island sensitivity of in-situ *wh*-DPs in Korean. Unlike in-situ

² We shall return to other possible explanations for this contrast in the General Discussions section.

wh-adverbials that mandatorily undergo covert *wh*-movement to their scope positions, the quantificational dependency of in-situ *wh*-DPs can be established through unselective binding without movement, and is thus immune to movement constraints (Pesetsky 1987; Nishigauchi 1990; Tsai 1994; Watanabe 2001).³ If the adjunct island effect observed in Experiment 1 reflects a discourse-level constraint on *wh*-questions with or without movement, we should expect the same island effect on *wh*-in-situ. If the temporal adjunct island effect does not constrain *wh*-in-situ, the island effect in Experiment 1 cannot be reduced to a discourse-level constraint in *wh*-questions, and instead points to genuine syntactic movement in *wh*-scrambling.

3 Experiment 2

Under certain discourse-level accounts of island effects (e.g., Erteschik-Shir 1973; Ambridge & Goldberg 2008; Goldberg 2013; Namboodiripad et al. 2022; Cuneo & Goldberg 2022; 2023; Lu et al. 2025), island effects (including the temporal adjunct island effect observed in Experiment 1) could reflect a discourse constraint on question formation that is orthogonal to whether syntactic movement is involved: discourse backgrounded/pre-supposed content cannot be questioned. Under these accounts, the existence of island effects cannot be taken as evidence for the existence of syntactic movement. Experiments 2 and 3 combined test whether this is a plausible explanation for the island effect observed in Experiment 1. Experiment 2 tests whether the temporal adjuncts used in Experiment 1 are indeed discourse backgrounded, as expected under a discourse account of island effect. Experiment 3 tests whether the temporal adjunct clauses restrict in-situ *wh*-questions where there is no overt syntactic movement. If Experiment 2 shows that the temporal adjunct clauses are interpreted as discourse backgrounded, but Experiment 3 shows that the temporal adjunct clauses do not restrict *wh*-in-situ, the results from the two experiments taken together would challenge the discourse account as an explanation for the island effect observed in Experiment 1 (alternative to an explanation where *wh*-scrambling involves genuine syntactic movement restricted by islands).

3.1 Methods, materials, and participants

In Experiment 2, following previous work (Goldberg 2013; Namboodiripad et al. 2022; Cuneo & Goldberg 2022; 2023; Fergus et al. 2025), we use the negation test to probe discourse backgroundedness. The negation test operates on the assumption that sentential negation negates at-issue contents, and does not affect backgrounded contents. For example, consider the English sentences (4a) and (4b). The polarity of the matrix clause does not affect the embedded content in the adjunct clause (i.e., that Jisoo met Hyunwoo). When asked *how likely is it that Jisoo met Hyunwoo* after reading these two sentences, listeners would give similar ratings, suggesting a high backgroundedness of the adjunct clause. By contrast, complement clauses like the ones in (5a) and (5b) in English are

³ Note that some studies claim that *wh*-DPs also undergo covert movement (Huang 1982), or that in-situ *wh*-DP questions involve overt operator movement that is subject to the subadjacency condition like overt movements (Watanabe 1992). Furthermore, some recent experimental studies also suggest that in-situ *wh*-DPs are sensitive to various island constraints, challenging the traditional view that they do not undergo covert movement (Tanaka 2025 for Japanese, Chieu 2022 for Vietnamese, and Lu et al. 2020, Kim et al. 2023, Chen 2024, and Yue & Xiang 2025 for Mandarin Chinese, cf. Tian et al. 2022; Jin & Yan 2025). To preview the findings in Experiment 3, our results suggest that at least in Korean, in-situ *wh*-DPs are not sensitive to the temporal adjunct island effect, in line with the analysis that in-situ *wh*-DPs do not need to undergo covert movement, but rather licensed through a non-island-sensitive way (e.g., unselective binding).

usually more at-issue than temporal adjunct clauses. Listeners would thus consider the embedded proposition (Jisoo met Hyunwoo) more likely true given sentence (5a) than given sentence (5b).

- (4) a. Minho laughed when Jisoo met Hyunwoo.
b. Minho didn't laugh when Jisoo met Hyunwoo.
- (5) a. Minho said that Jisoo met Hyunwoo.
b. Minho didn't say that Jisoo met Hyunwoo.

This negation test has been extensively used to test for the correlation between discourse backgroundedness and islandhood in English. In this experiment, we adopt the same negation test to test whether temporal adjunct clauses are backgrounded in Korean.

A total of 44 native speakers of Korean were recruited for the experiment and compensated \$3 for their participation. Participants who took part in Experiment 1 were not eligible to participate in the current experiment. They were presented with context sentences containing adjunct or complement clauses, accompanied by a question probing how likely they think the embedded proposition is true. We adopt a 2-by-2 factorial design manipulating POLARITY (negated vs. not negated) and STRUCTURE (adjunct vs. complement), yielding 4 conditions for the context sentences. The adjunct/not negated sentences are the declarative forms of the island condition sentences from Experiment 1, and the complement/not negated sentences are the declarative forms of the non-island condition sentences from Experiment 1. The negation condition sentences are the negated forms of their not-negated counterparts. Example stimuli are shown in (6). The same 16 lexicalizations as in Experiment 1 are used. After reading each context-question pair, participants were asked to answer the question on a 1-7 Likert scale, with 1 representing "not likely at all", and 7 representing "very likely".

(6) Experiment 2 sample stimuli

a. *Negated, Adjunct*

Minho-ka Jisoo-ka Hyunwoo-lul mannassul-ttay wusci-anh-ass-ta.
Minho-NOM Jisoo-NOM Hyunwoo-ACC meet.PST.ADN-when laugh-NEG-PST-DECL
'Minho did not laugh when Jisoo met Hyunwoo.'
cilmwun: Jisoo-ka Hyunwoo-lul mannasseul kanungseng-i elmana toynayo?
Question: Jisoo-NOM Hyunwoo-ACC meet.PST possibility-NOM how.much be.Q?
'Question: How likely is it that Jisoo met Hyunwoo?'

b. *Negated, Complement*

Minho-ka Jisoo-ka Hyunwoo-lul mannass-tako tutci-anh-ass-ta.
Minho-NOM Jisoo-NOM Hyunwoo-ACC meet.PST-COMP hear-NEG-PST-DECL
'Minho did not hear that Jisoo met Hyunwoo.'
cilmwun: Jisoo-ka Hyunwoo-lul mannasseul kanungseng-i elmana toynayo?
Question: Jisoo-NOM Hyunwoo-ACC meet-PST possibility-NOM how.much be.Q?
'Question: How likely is it that Jisoo met Hyunwoo?'

c. *Not negated, Adjunct*

Minho-ka Jisoo-ka Hyunwoo-lul mannasseul-ttay wus-ess-ta.
Minho.NOM Jisoo-NOM Hyunwoo-ACC meet.PST.ADN-when laugh-PST-DECL
'Minho laughed when Jisoo met Hyunwoo.'
cilmwun: Jisoo-ka Hyunwoo-lul mannasseul kanungseng-i elmana toynayo?
Question: Jisoo-NOM Hyunwoo-ACC meet.PST possibility-NOM how.much be.Q?
'Question: How likely is it that Jisoo met Hyunwoo?'

d. *Not negated, Complement*

Minho-ka Jisoo-ka Hyunwoo-lul mannass-ta-ko tul-ess-ta.

Minho-NOM Jisoo-NOM Hyunwoo-ACC meet.PST-COMP hear-PST-DECL

‘Minho heard that Jisoo met Hyunwoo.’

cilmwun: Jisoo-ka Hyunwoo-lul mannasseul kanungseng-i elmana toynayo?

Question: Jisoo-NOM Hyunwoo-ACC meet.PST possibility-NOM how.much be.Q?

‘Question: How likely is it that Jisoo met Hyunwoo?’

3.2 Results and discussion

The negation test responses are shown in Figure 2. We fit an ordinal regression model predicting the responses with the sum-coded fixed effects of POLARITY, STRUCTURE, and their interaction, as well as the maximal by-item and by-participant random effect structure. There is a significant main effect of POLARITY ($\beta = 1.01$, $SE = 0.14$, $z = 7.35$, $p < 0.001$) where the not-negated conditions received higher score than the negated conditions. There is a significant main effect of STRUCTURE ($\beta = -1.85$, $SE = 0.21$, $z = -8.92$, $p < 0.001$), where the complement conditions received lower scores than the adjunct conditions. There is also a significant POLARITY $\times$ STRUCTURE interaction ($\beta = 0.99$, $SE = 0.14$, $z = 6.86$, $p < 0.001$), where the POLARITY contrast is greater for the complement conditions than for the adjunct conditions. This significant interaction effect shows that the -ttay temporal adjunct clauses used in the stimuli of Experiment 1 are indeed interpreted as discourse backgrounded by the participants.

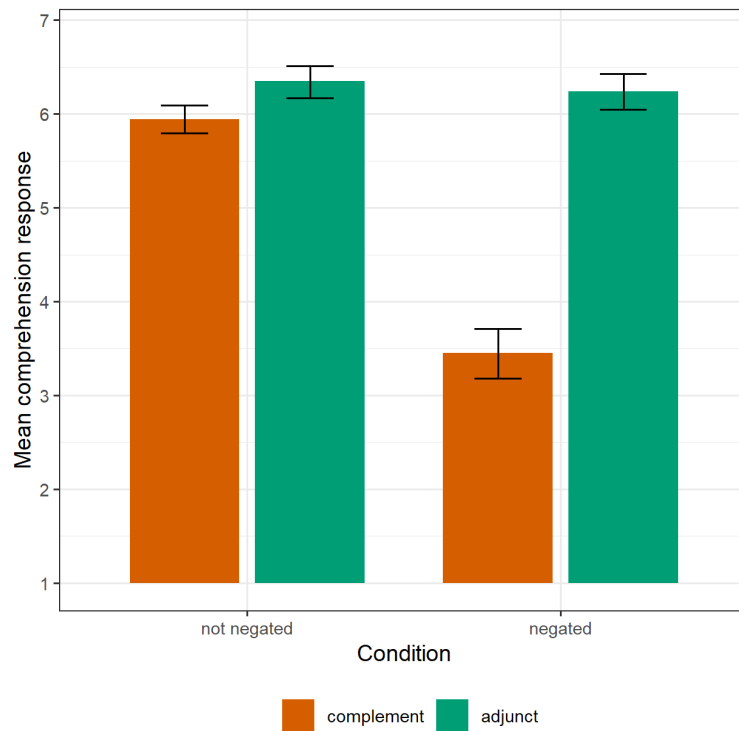


Figure 2: Negation test responses for Experiment 2

4 Experiment 3

In Experiment 2, we have shown that the *-ttay* temporal adjunct clauses in Korean, just like their English counterparts, are interpreted to be discourse backgrounded. In this experiment, we test whether in-situ *wh*-DPs are sensitive to the *-ttay* temporal adjunct island in Korean. If the adjunct island effect observed in Experiment 1 reflects a discourse constraint on *wh*-questions regardless of whether movement is involved, we expect the same island effect to show up for *wh*-in-situ sentences, given the discourse backgrounded status of adjuncts. If the island effect in Experiment 1 reflects a locality constraint on movement, we do not expect in-situ *wh*-DPs to show the same effect.

4.1 Methods, materials, and participants

Experiment 3 adopts the same design and stimuli as Experiment 1, except that the *wh*-elements stay in-situ. The “long” and “short” conditions of the LENGTH manipulation thus refers to the dependency distance between the *wh*-in-situ and their scope position. Example stimuli are shown in (7).

Note that this experiment uses in-situ *wh*-DPs in the stimuli, as opposed to *wh*-adverbials. There is a long-documented generalization that in *wh*-in-situ languages, *wh*-adverbials are island sensitive while *wh*-DPs are not, presumably because in-situ *wh*-DPs can establish quantification dependency with their scope positions without undergoing covert *wh*-movement (Tsai 1994; Watanabe 2001; Nishigauchi 1990; Pesetsky 1987). *Wh*-DPs therefore provide a great test for a discourse-based explanation for the temporal adjunct island effect we observed in Experiment 1: if the effect in Experiment 1 reflects a discourse constraint on question formation, it should apply to in-situ *wh*-questions as well even when there is no syntactic movement.

A total of 76 native speakers of Korean were recruited for the experiment and were compensated approximately \$3 for their participation. Participants who took part in Experiments 1 or 2 were not eligible to participate in the current experiment. We further applied the same exclusion criterion based on the filler responses as in Experiment 1, but no participant was excluded.

(7) Experiment 3 sample stimuli sentences

a. *Island, Long*

cilmwun: Minho-nun Jisoo-ka nwukwu-lul mannal-ttay wus-ess-ni?
 Question: Minho-TOP Jisoo-NOM who-ACC meet.PST.ADN-when laugh-PST-Q?
 tap: Hyunwoo
 Answer: Hyunwoo
 ‘Q: Who did Minho laugh when Jisoo met? A: Hyunwoo.’

b. *Island, Short*

cilmwun: Nwu-ka Jisoo-ka Hyunwoo-lul mannassul-ttay wus-ess-ni?
 Question: Who-NOM Jisoo-NOM Hyunwoo-ACC meet.PST.ADN-when laugh-PST-Q?
 tap: Minho
 Answer: Minho
 ‘Q: Who laughed when Jisoo met Hyunwoo? A: Minho.’

c. *Non-island, Long*

cilmwun: Minho-nun Jisoo-ka nwukwu-lul mannass-tako tul-ess-ni?
 Question: Minho-TOP Jisoo-NOM who-ACC meet.PST-COMP hear-PST-Q?

- tap: Hyunwoo
 Answer: Hyunwoo
 ‘Q: Who did Minho hear that Jisoo met? A: Hyunwoo.’
- d. *Non-island, Short*
 cilmwun: Nwu-ka Jisoo-ka Hyunwoo-lul mannass-tako tul-ess-ni?
 Question: Who-NOM Jisoo-NOM Hyunwoo-ACC meet.PST-COMP hear-PST-Q?
 tap: Minho
 Answer: Minho
 ‘Q: Who heard that Jisoo met Hyunwoo? A: Minho’

4.2 Results and discussion

The mean acceptability rating for each condition is shown in Figure 3. We fit an ordinal regression model predicting the acceptability rating with the sum-coded fixed effects of STRUCTURE, LENGTH, and their interaction. Also included were by-item and by-participant random intercepts and random slopes for the fixed effects and their interaction. There is a significant main effect of LENGTH ($\beta = 0.71$, $SE = 0.12$, $z = 5.93$, $p < 0.001$) where the short conditions are rated higher than the long conditions. There is no significant effect of STRUCTURE ($\beta = -0.072$, $SE = 0.078$, $z = -0.93$, $p = 0.35$) or STRUCTURE $\times$ LENGTH interaction ($\beta = -0.014$, $SE = 0.080$, $z = -0.18$, $p = 0.86$).

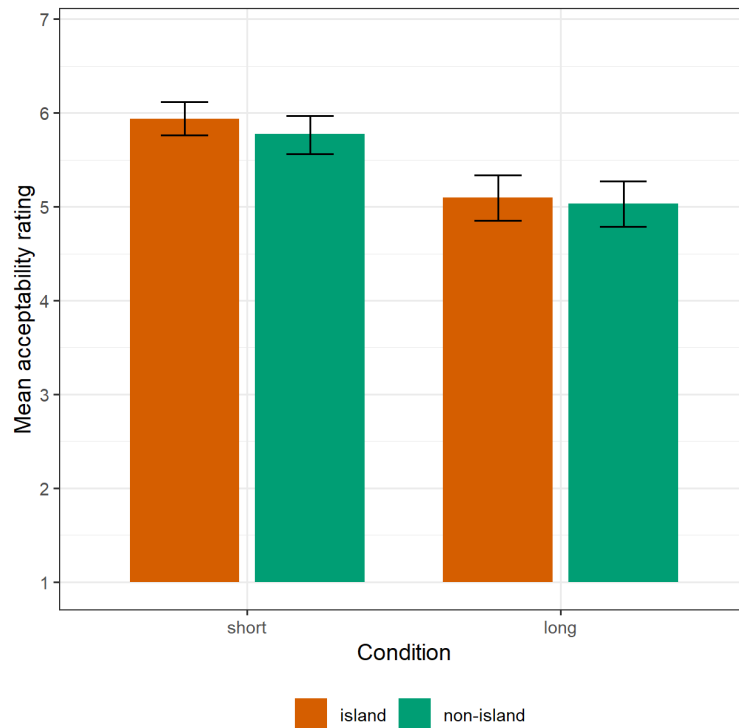


Figure 3: Acceptability ratings for Experiment 3

The results suggest that there is no evidence for any temporal adjunct island effect on in-situ *wh*-DPs, replicating previous findings in the literature (Kim & Goodall 2016). The results also point to a dissociation between discourse backgroundedness and islandhood in Korean: even though clausal adjuncts are discourse backgrounded as shown in Experiment 2, forming *wh*-questions targeting elements embedded in clausal adjuncts is

not necessarily degraded. Therefore, the *-ttay* temporal adjunct island effect observed in Experiment 1 cannot be reduced to a discourse constraint on question formation, and instead should reflect a genuine syntactic island constraint on movement.

5 General discussion

This squib makes three main contributions to the study of scrambling and island effects in Korean.

First, on the empirical side, we present experimental evidence suggesting that *wh*-scrambling is sensitive to the *-ttay* temporal adjunct island in Korean. This contrasts with earlier work demonstrating adjunct island insensitivity for the long-distance scrambling of non-*wh* DPs (Ko et al. 2019; Kim 2026), as well as with Kim & Goodall's 2016 and our findings in Experiment 3 that in-situ *wh*-DPs are also insensitive to the adjunct island. Our finding thus expands the empirical landscape of long-distance scrambling and the adjunct island effect in Korean. Admittedly, we only examined one particular type of clausal adjunct (*-ttay* clauses), and it is possible that movements may display different sensitivity to different types of clausal adjuncts, as have been demonstrated crosslinguistically (Truswell 2007; Bondevik et al. 2021; Bondevik & Lohndal 2023). We leave it to future studies to further expand the empirical coverage by testing other types of adjunct clauses.

Second, our findings point to the theoretical possibility that *wh*-scrambling, but not regular scrambling, may be derived through syntactic movement. If *wh*-scrambling necessarily proceeds via overt A'-movement, the adjunct island effect follows naturally. Meanwhile, the island insensitivity of non-*wh* scrambling remains consistent with base-generation accounts of scrambling (e.g. Bošković & Takahashi 1998, Fanselow 2001) in which scrambled DPs have the option to be directly merged at their surface positions. An open question, however, is what bars *wh*-elements from being base-generated at its surface scrambled position. We do not intend to offer a solution in this squib, and will leave this as an important question for future work.

Apart from the aforementioned explanation based on the different derivation paths for *wh*- and non-*wh*-scrambling, there are a few alternative explanations for why our findings diverge from previous experiments on non-*wh*-scrambling in Korean. In Ko et al. (2019), the authors gave two speculative explanations for why they didn't find island effects on non-*wh*-scrambling. First, the authors speculate that Korean allows multiple specifiers, and the scrambled phrases in Korean can escape islands by merging as multiple specifiers of C. This explanation is implausible: the availability of multiple specifier positions might account for the lack of *wh*-island effects in Korean (which is also reported in Ko et al. 2019), but not clausal adjunct island effects, which cannot be straightforwardly attributed to the lack of a CP specifier position at the island edge. A second explanation provided in Ko et al. (2019) is that their experiment might be confounded by a floor effect, which masks over an island effect by compressing the acceptability contrasts near the bottom of the scale. However, Kim (2026) replicates Ko et al.'s 2019 lack of adjunct island effect, and the average acceptability Z-scores for all conditions in Kim (2026) are positive and thus not confounded by any floor effect. Finally, a third possible explanation is that the short dependency length conditions in our Experiment 1 are different from the short conditions in Ko et al. (2019), which might have led to different experimental results. Our stimuli involve subject *wh*-elements in the short conditions and object *wh*-elements in the long condition, whereas Ko et al.'s stimuli involve object *wh*-elements in all conditions. Ko et al. observed a pronounced acceptability contrast between their

SHORT.ISLAND and SHORT.NON-ISLAND conditions, and this contrast drives a statistically significant sub-additive interaction effect between LENGTH and STRUCTURE, in the opposite direction of an island effect. However, in our Experiment 1, we did not observe any pronounced contrast between the short conditions. One might speculate that the difference in findings of the two studies is driven by the difference in the short conditions. An anonymous reviewer suggests that the contrast between the two studies might reflect a subject-object asymmetry in scrambling in Korean. However, any such asymmetry should be reflected as a main effect in DEPENDENCY LENGTH in Experiment 1, and should not affect the super-additive island effect. Alternatively, there could be a penalty that selectively affects object scrambling from the matrix when an adjunct clause is present. Since both short conditions in Ko et al. (2019) involve the extraction of the indirect object DP from a position that linearly follows the embedded (complement or adjunct) clause, the finding in Ko et al. (2019) could reflect a processing-level penalty that burdens the processing of dependencies linearly spanning adjunct clauses but not ones that linearly span complement clauses.⁴ Nevertheless, one should note that Kim (2026) used stimuli similar to the ones in Ko et al. (2019) where all scrambled elements are objects, but there is no contrast between the two short conditions in Kim (2026). We leave it to future studies to test whether this contrast in the short conditions reported in Ko et al. (2019) is a reliably replicable effect.

A third contribution of this study is that the results of Experiments 2 and 3 inform the broader debates about the status of discourse-based accounts of island effects. Experiment 2 confirmed that the *-ttay* adjuncts in Korean are indeed interpreted as discourse backgrounded, just like their English counterparts. However, Experiment 3 showed that *wh*-in-situ questions can target elements inside these *-ttay* clauses without any island effect. This means that the degradation observed in Experiment 1 cannot be reduced to a pragmatic clash between the new-information-seeking nature of *wh*-questions and the backgrounded status of clausal adjuncts. On a higher level, given the dissociation between discourse backgroundedness and islandhood we observed, the findings of Experiments 2 and 3 constitute a counter-example to the discourse-based accounts that attribute island effects to the clash between the discourse foregrounding/new-information-seeking function of *wh*-questions, and the discourse backgrounded status of certain structural domains (Ambridge & Goldberg 2008; Goldberg 2013; Nambodiripad et al. 2022; Cuneo & Goldberg 2022). Nevertheless, this does not mean that all discourse-based accounts of islands are untenable. One recent proposal by Winckel et al. (2025) states that only *wh*-elements that undergo movement are foregrounded. Our findings are still compatible with this particular account: the island effect in Exp.1 arises because the *wh*-scrambling construction involves movement of the *wh*-element, whereas the in-situ argument *wh*-questions in Exp.3 do not involve (overt or covert) movement and thus do not give rise to any discourse clash. Note that under this account, we can still conclude that *wh*-scrambling in Korean involves movement: if the sentence-initial *wh*-elements are base-generated, Winckel et al.'s (2025) account would predict no island effect since no movement is involved, contrary to our finding.

In sum, this squib presents a data point that *wh*-scrambling, unlike non-*wh*-scrambling or *wh*-in-situ, is sensitive to the adjunct island constraint in Korean. This result is compatible with a movement-based analysis of *wh*-scrambling, and invites a reevaluation of the assumption that Korean scrambling constitutes a single unified operation.

⁴ This penalty should not be syntactic, since the two conditions involve the same syntactic movement from the same position, and the embedded complement and adjunct clause boundaries are not structurally crossed by the movements.

Data availability statement

All stimuli, data files, and analysis scripts can be accessed at <https://osf.io/bfwtu/>



Competing interests

The authors have no competing interests to declare.

Ethics and consent

The experiments included in this study have been approved by the Sungkyunkwan University Institutional Review Board, approval number 2021-04-009, and the Northwestern University Institutional Review Board, approval number STU00224884.

Funding information

This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2025S1A5A8008853).

Acknowledgement

The authors are grateful to the three anonymous reviewers at Glossa, Hee Joong Choi, Dongwoo Park, and Nozomi Tanaka for their valuable comments at various stages of this project. All remaining errors are ours.

References

- Abeillé, Anne & Hemforth, Barbara & Winckel, Elodie & Gibson, Edward. 2020. Extraction from subjects: Differences in acceptability depend on the discourse function of the construction. *Cognition* 204. 104293.
- Ambridge, Ben & Goldberg, Adele E. 2008. The island status of clausal complements: Evidence in favor of an information structure explanation. *Cognitive Linguistics* 19(3). 357–389.
- Bailyn, John Frederick. 2001. On scrambling: A reply to Bošković and Takahashi. *Linguistic inquiry* 32(4). 635–658.
- Bondevik, Ingrid & Kush, Dave & Lohndal, Terje. 2021. Variation in adjunct islands: The case of Norwegian. *Nordic Journal of Linguistics* 44(3). 223–254.
- Bondevik, Ingrid & Lohndal, Terje. 2023. Extraction from finite adjunct clauses: an investigation of relative clause dependencies in Norwegian. *Glossa: a journal of general linguistics* 8(1).
- Bošković, Željko & Takahashi, Daiko. 1998. Scrambling and last resort. *Linguistic inquiry* 29(3). 347–366.
- Çakır, Sinan. 2016. Island constraints and adjunct & argument asymmetry in Turkish. *Dilbilim Araştırmaları Dergisi* 27(2). 1–15.
- Çakır, Sinan. 2017. The wh-adverbial & which-NP construction asymmetry within island structures in Turkish. *Dil ve Dilbilimi Çalışmaları Dergisi* 13(1). 232–243.
- Chen, Xu. 2024. *Wh-island effects in Chinese: A formal experimental study*. John Benjamins Publishing Company.

- Chieu, Quynh Kim. 2022. *Syntactic islands and wh-in-situ languages: the case of vietnamese*. Radboud University Bachelor's thesis.
- Cho, Juyeon. 2018. Korean wh-island effects in scrambling constructions. *Japanese/Korean Linguistics* 25. 1–12.
- Chomsky, Noam & Lasnik, Howard. 1977. Filters and control. *Linguistic inquiry* 8(3). 425–504.
- Cuneo, Nicole & Goldberg, Adele E. 2022. Islands effects without extraction: the discourse functions of constructions predicts island status. In *Proceedings of the 44th annual meeting of the cognitive science society*, vol. 44.
- Cuneo, Nicole & Goldberg, Adele E. 2023. The discourse functions of grammatical constructions explain an enduring syntactic puzzle. *PsyArXiv*.
- Erteschik-Shir, Nomi. 1973. *On the nature of island constraints*. Massachusetts Institute of Technology dissertation.
- Fanselow, Gisbert. 2001. Features, θ -roles, and free constituent order. *Linguistic inquiry* 32(3). 405–437.
- Fergus, Abigail & Belluck, Arielle & Cuneo, Nicole & Goldberg, Adele. 2025. Islands result from clash of functions: Single-conjunct wh-qs. In *Proceedings of the annual meeting of the cognitive science society*, vol. 47.
- Fukui, Naoki. 1993. Parameters and optionality. *Linguistic inquiry* 24(3). 399–420.
- Goldberg, Adele E. 2013. Backgrounded constituents cannot be “extracted”. In Sprouse, Jon & Hornstein, Norbert (eds.), *Experimental syntax and island effects*, 221–238. Cambridge: Cambridge University Press.
- Han, Chung-hye. 2013. On the syntax of relative clauses in korean. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 58(2). 319–347.
- Han, Chung-hye & Kim, Jong-Bok. 2004. Are there “double relative clauses” in korean? *Linguistic Inquiry* 315–337.
- Han, Jongim. 1992. Syntactic movement analysis of korean relativization. 어학연구.
- Hoji, Hajime. 1985. *Logical form constraints and configurational structures in japanese*. University of Washington.
- Huang, C.-T. James. 1982. *Logical relations in Chinese and the theory of grammar*. Massachusetts Institute of Technology dissertation. Published by Garland, New York, 1998.
- Jin, Dawei & Yan, Hanbo. 2025. Reassessing the argument–adjunct asymmetry in wh-in-situ islands in mandarin: An experimental investigation. *Natural Language & Linguistic Theory* 43(3). 1327–1363.
- Kim, Boyoung & Goodall, Grant. 2016. Islands and non-islands in native and heritage korean. *Frontiers in psychology* 7. 134.
- Kim, Ilkyu. 2013. Rethinking “island effects” in korean relativization. *Language Sciences* 38. 59–82.
- Kim, Ilkyu. 2026. Extraction from adjunct and relative clause islands in korean. *Linguistic Research* 43(1). 61–103.
- Kim, Ilkyu & Ji, Yunjoo. 2023. On the nature of enpc effects in korean scrambling constructions. *Linguistic Research* 40(2). 151–181.
- Kim, Jong-Bok. 2019. Predicate topicalization in korean: A construction-based hpcg approach. 언어 44(3). 395–423.
- Kim, Nayoun & Li, Ziyang & Lu, Jiayi. 2023. Island-sensitivity of two different interpretations of why in Chinese. *Frontiers in Psychology* 13. 1059823. <https://doi.org/10.3389/fpsyg.2022.1059823>

- Ko, Heejeong. 2007. Asymmetries in scrambling and cyclic linearization. *Linguistic inquiry* 38(1). 49–83.
- Ko, Heejeong. 2014. *Edges in syntax: Scrambling and cyclic linearization*, vol. 50. Oxford University Press.
- Ko, Heejeong & Chung, Han-byul & Kim, Kitaek & Sprouse, Jon. 2019. An experimental study on scrambling out of islands: To the left and to the right. *Language & Information Society* 37(null). 287–323.
- Kornfilt, Jaklin & Bayer, Josef. 1994. Against scrambling as an instance of move-alpha. ~~*Studies on scrambling: Movement and non-movement approaches to free word order phenomena*~~ 41. 17–60.
- Kuroda, Shige-Yuki. 1988. Whether we agree or not: A comparative syntax of english and japanese. *Linguisticae Investigationes* 12(1). 1–47.
- Kush, Dave & Lohndal, Terje & Sprouse, Jon. 2019. On the island sensitivity of topicalization in norwegian: An experimental investigation. *Language* 95(3). 393–420.
- Lee, Young-Suk. 1993. *Scrambling as case-driven obligatory movement*. University of Pennsylvania.
- Lu, Jiayi & Pan, Dingyi & Degen, Judith. 2025. Evidence for a discourse account of manner-of-speaking islands. *Language* 101(4). 627–659.
- Lu, Jiayi & Thompson, Cynthia K & Yoshida, Masaya. 2020. Chinese wh-in-situ and islands: A formal judgment study. *Linguistic Inquiry* 51(3). 611–623.
- Miyagawa, Shigeru. 1997. Against optional scrambling. *Linguistic inquiry* 1–25.
- Na, Younghee & Huck, Geoffrey J. 1993. On the status of certain island violations in korean. *Linguistics and Philosophy* 16(2). 181–229.
- Namboodiripad, Savithry & Cuneo, Nicole & Kramer, Mathew A & Sedarous, Yourdanis & Bisnath, Felicia & Sugimoto, Yushi & Goldberg, Adele. 2022. Backgroundedness predicts island status of non-finite adjuncts in english. In *Proceedings of the 44th annual meeting of the cognitive science society*, vol. 44.
- Neeleman, Ad. 1994. Scrambling as a d-structure phenomenon. ~~*Studies on scrambling: Movement and non-movement approaches to free word order phenomena*~~ 387. 430.
- Nishigauchi, Taisuke. 1990. Construing wh. In *Quantification in the theory of grammar*, 116–176. Springer.
- Pesetsky, David. 1987. Wh-in-situ: Movement and unselective binding. In Reuland, Eric & ter Meulen, Alice (eds.), *The representation of (in)definiteness*, 98–129. Cambridge, MA: The MIT Press.
- Ross, John R. 1967. *Constraints on variables in syntax*. Massachusetts Institute of Technology dissertation.
- Saito, Mamoru. 1982. Scrambling, topicalization and strong crossover. ms.
- Saito, Mamoru. 1985. *Some asymmetries in japanese and their theoretical implications*. Massachusetts Institute of Technology dissertation.
- Sprouse, Jon & Caponigro, Ivano & Greco, Ciro & Cecchetto, Carlo. 2016. Experimental syntax and the variation of island effects in english and italian. *Natural Language & Linguistic Theory* 34(1). 307–344.
- Sprouse, Jon & Fukuda, Shin & Ono, Hajime & Kluender, Robert. 2011. Reverse island effects and the backward search for a licenser in multiple wh-questions. *Syntax* 14(2). 179–203.
- Sprouse, Jon & Wagers, Matt & Phillips, Colin. 2012. A test of the relation between working-memory capacity and syntactic island effects. *Language* 88(1). 82–123.
- Tanaka, Nozomi. 2025. Subliminal island effects in Japanese complex NPs with argument wh-in-situ. *Linguistic Inquiry* 1–24. Advance online publication.

- Tian, Qilin & Park, Myung-Kwan & Yang, Xiaodong. 2022. Mandarin *chinese* wh-in-situ argument–adjunct asymmetry in island sensitivity: Evidence from a formal judgment study. *Frontiers in Psychology* 13. 954175.
- Truswell, Robert. 2007. Extraction from adjuncts and the structure of events. *Lingua* 117(8). 1355–1377.
- Tsai, Cheng-Yu Edwin. 2009. *Wh-dependency in vietnamese and the syntax of wh-in-situ*. National Tsing Hua University MA thesis.
- Tsai, Wei-Tien Dylan. 1994. On nominal islands and LF extraction in Chinese. *Natural Language & Linguistic Theory* 12(1). 121–175.
- Watanabe, Akira. 1992. Subjacency and s-structure movement of wh-in-situ. *Journal of east Asian linguistics* 1(3). 255–291.
- Watanabe, Akira. 2001. Wh-in-situ languages. *The handbook of contemporary syntactic theory* 203–225.
- Winckel, Elodie & Abeillé, Anne & Hemforth, Barbara & Gibson, Edward. 2025. Discourse-based constraints on long-distance dependencies generalize across constructions in *english* and *french*. *Cognition* 254. 105950.
- Yue, Jiayuan & Xiang, Ming. 2025. The argument-adjunct asymmetry revisited: The role of focus alternatives in island effects. In Clemens, Lauren & Gribanova, Vera & Scontras, Gregory (eds.), *Syntax in uncharted territories: Essays in honor of maria polinsky*, University of California.